Flash the SUPER bootloader (Must read)

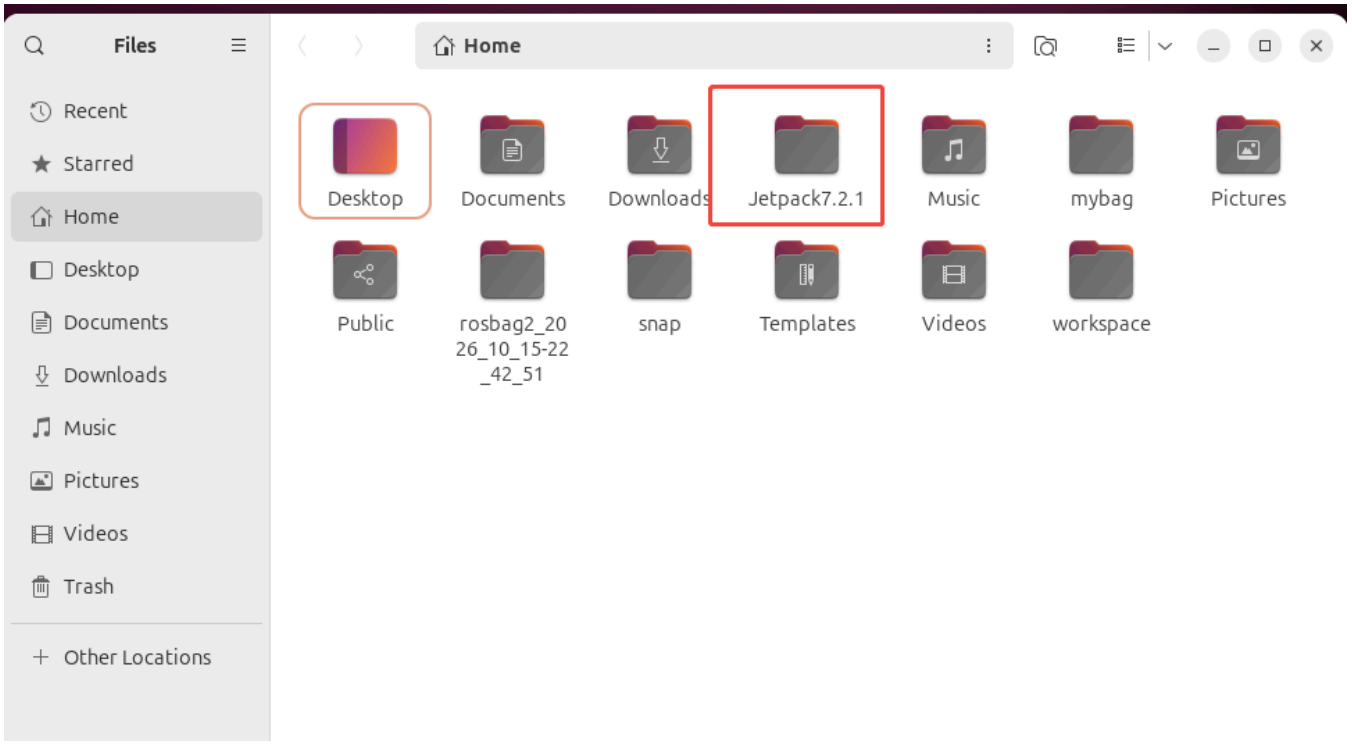
Flash the SUPER bootloader (Must read)

1. Flashing mode
    1.1. Hardware connection
    1.2. Software connection
2. Flash the bootloader
3. Start the system

Put the provided `Jetpack7.2.1` folder into the home directory of the Ubuntu virtual machine user. Taking the `yahboom` user as an example, the folder path should be:

```
/home/yahboom/Jetpack7.2.1
```

This tutorial starts from the prepared bundled folder, so there is no need to download and extract the BSP and the RootFS again. The bundled folder should already have the environment prepared and the QSPI generation script fixed; the one-click command of this tutorial assumes that `yahboom_flash.sh.x` is already provided in it.



If the bootloader of the mainboard is not the version corresponding to JetPack 7.2.1 and you want to use it together with our JetPack 7.2.1 system, before the first use you have to update the bootloader of the mainboard to **L4T R39.2.1**, and then install the bundled SSD and boot.

The flashing process only updates the bootloader and the UEFI in the QSPI of the mainboard; it does not write the SSD system. **No SSD has to be installed while flashing; it is recommended to remove the SSD while the board is powered off.**

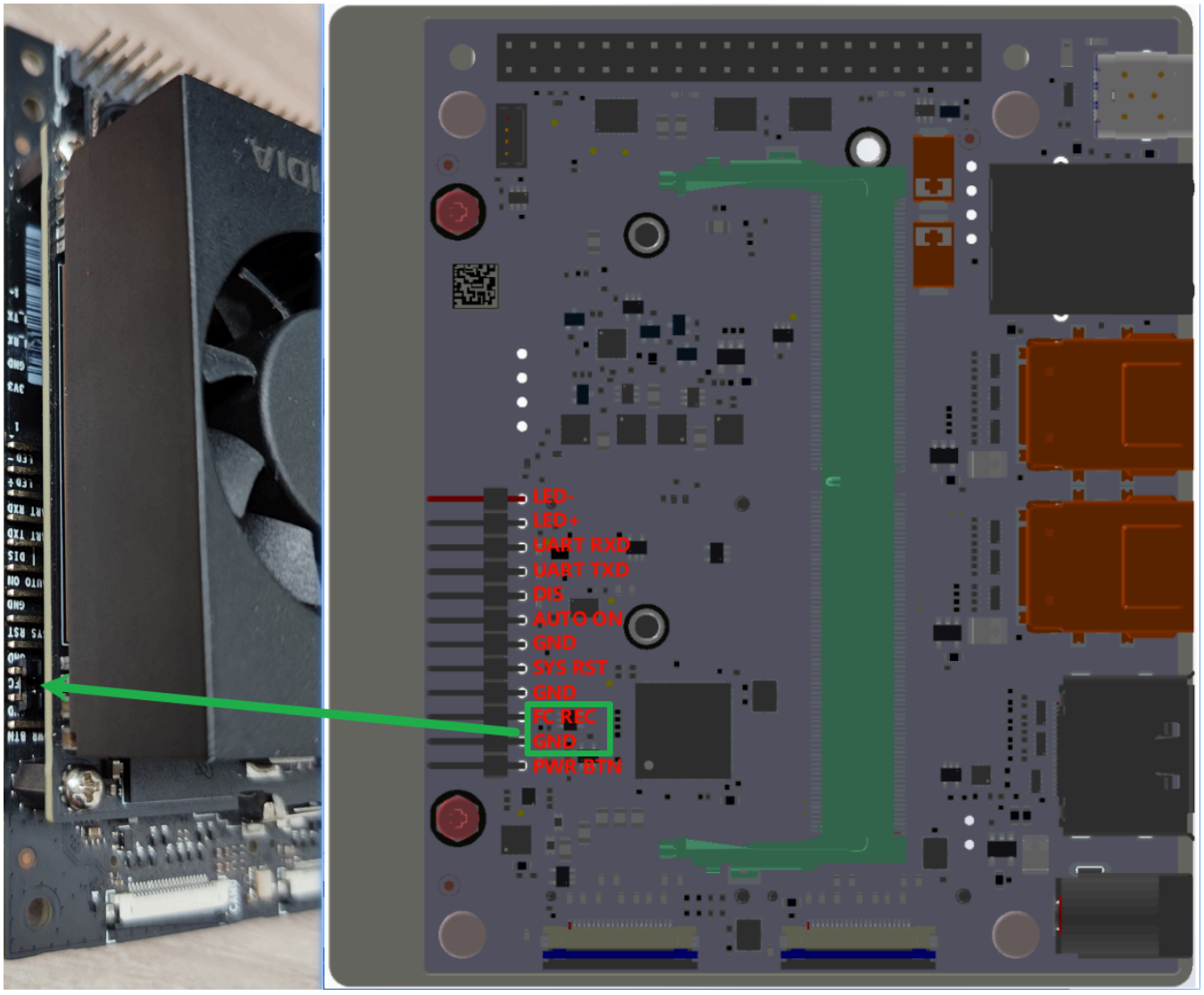
This tutorial follows the NVIDIA official Orin NX developer kit with the Super configuration, and the virtual machine used for flashing runs an x86_64 system with Ubuntu 22.04 or 24.04. For third-party carrier boards such as SUB and for other module models, first confirm whether the bundled files and the board configuration apply.

> Official flashing link: https://docs.nvidia.com/jetson/archives/r39.2.1/DeveloperGuide/SD/FlashingSupport.html#qspi-only-flashing

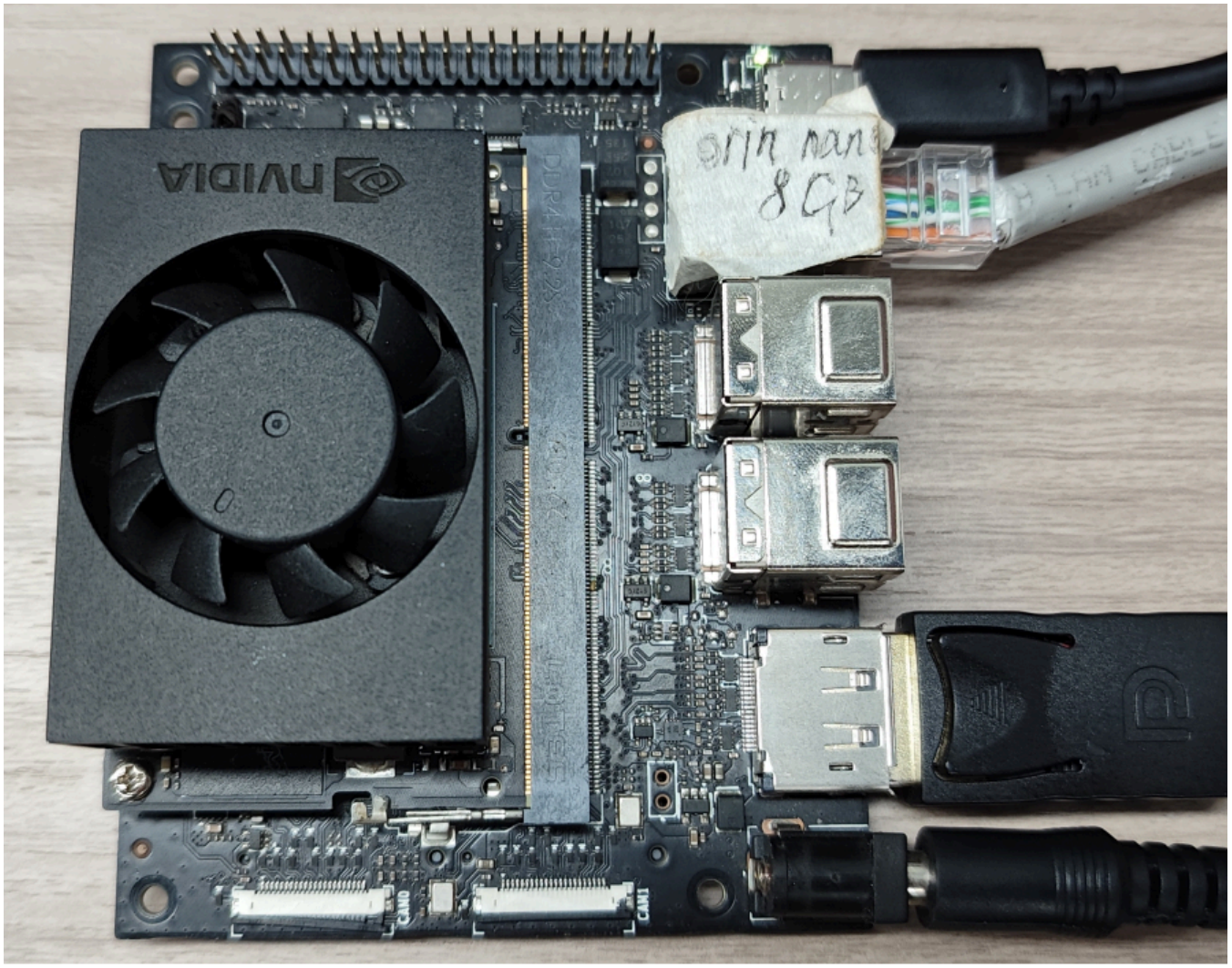
1. Flashing mode

1.1. Hardware connection

1. Disconnect the power of the Jetson mainboard and remove the SSD.
2. Following the pin markings of the carrier board, use a jumper cap to short `FC REC` and `GND`. The relevant pins of the official Orin NX carrier board are located below the core board, so the core board does not have to be removed.



3. Use a Type-C data cable that supports data transfer to connect the Jetson mainboard to the computer, and then connect the DC power adapter to power up the mainboard so that it enters Recovery mode.

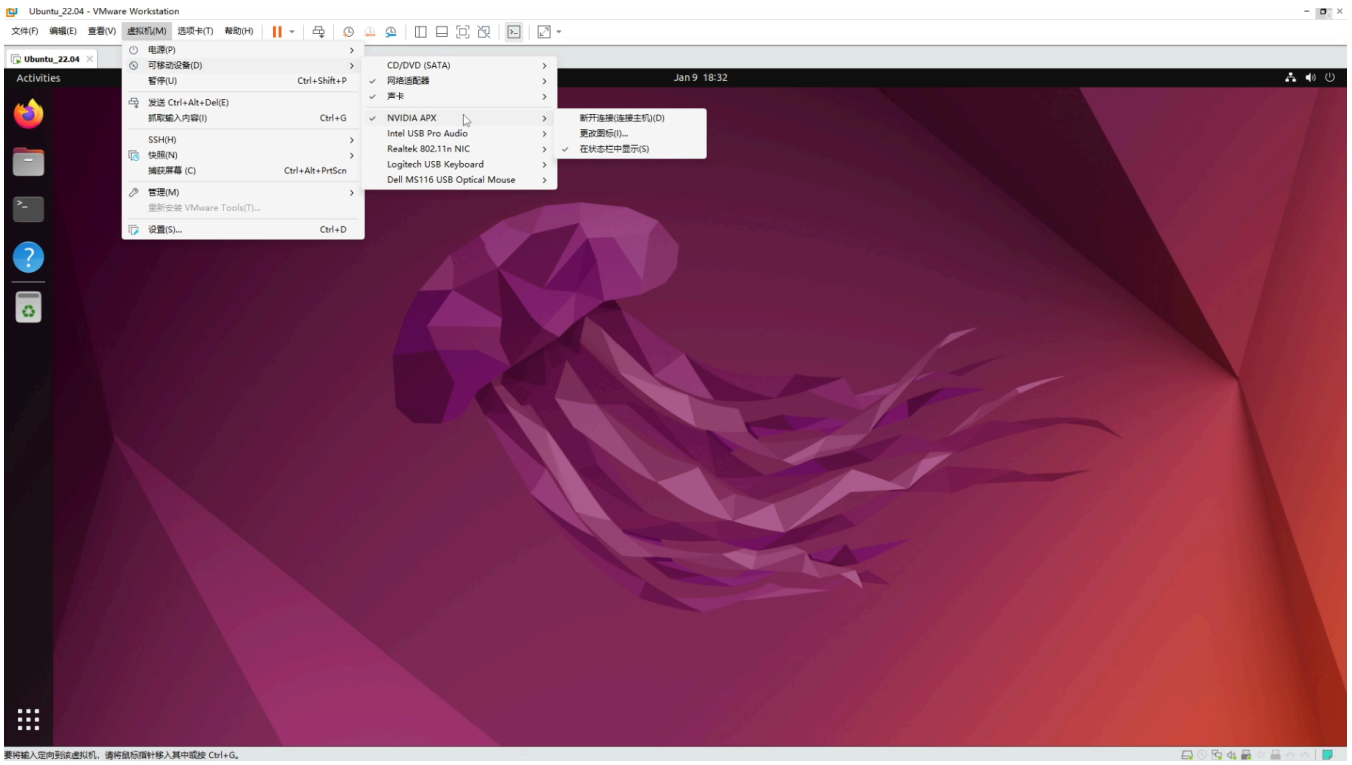


Note: Flashing the bootloader does not require the DP data cable and the network cable; a display is needed later when you check the boot screen. Keep the mainboard powered and the data cable connected while flashing.

1.2. Software connection

Start the Ubuntu virtual machine, find the NVIDIA USB device under "Virtual Machine → Removable Devices" in VMware and connect it to the virtual machine.

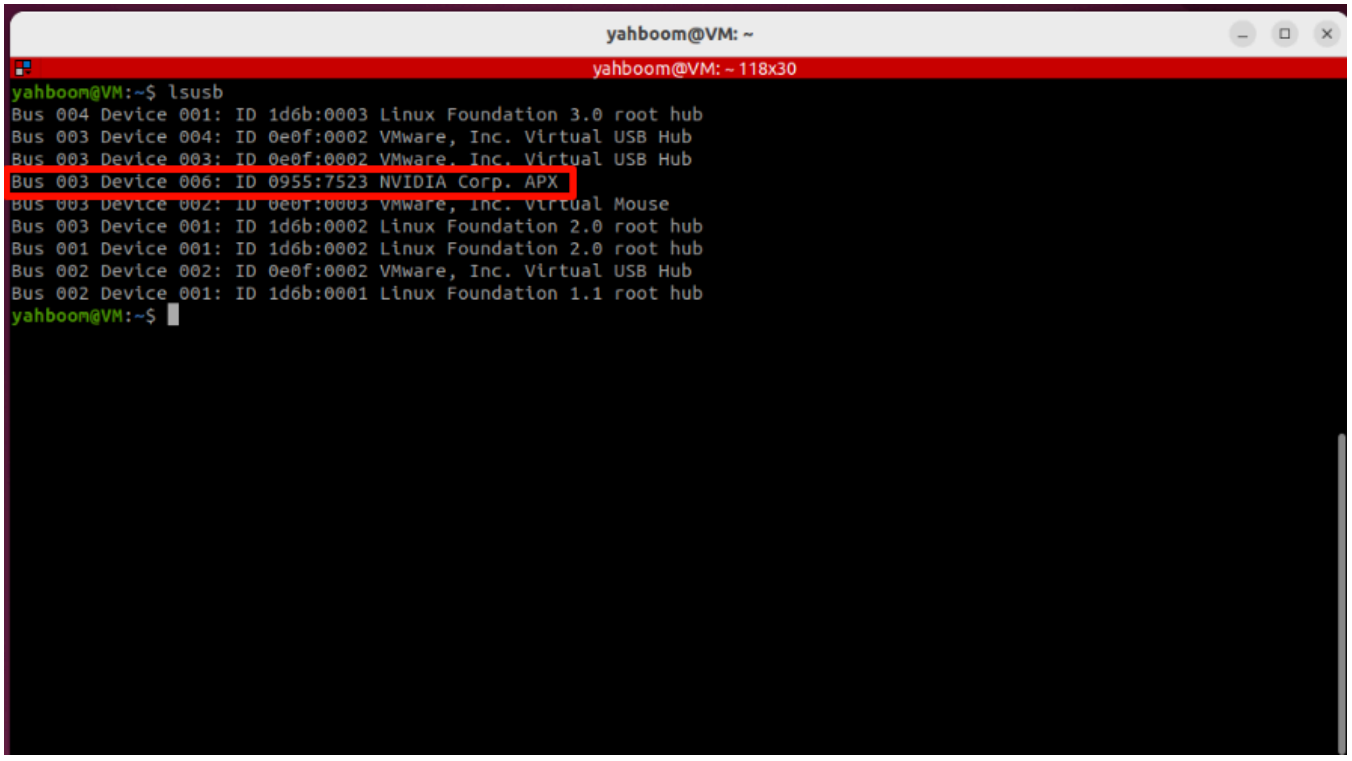
The device name may be shown as `NVIDIA Corp. APX`. After you select the connection, the device should be used by the Ubuntu virtual machine.



Open a terminal on the virtual machine desktop and run:

```
lsusb
```

Confirm that the NVIDIA device appears in the list, with the vendor ID `0955`. If it is not detected, first check the jumper cap, the power-up sequence, the Type-C data cable and the VMware USB connection state, and continue once they are confirmed.



2. Flash the bootloader

Open a terminal in the virtual machine as a normal user. The user name and the password of the bundled virtual machine are both `yahboom`; if you use your own Ubuntu system, use your own account and make sure that the flashing runtime dependencies are installed.

Enter the folder and confirm that the flashing script exists:

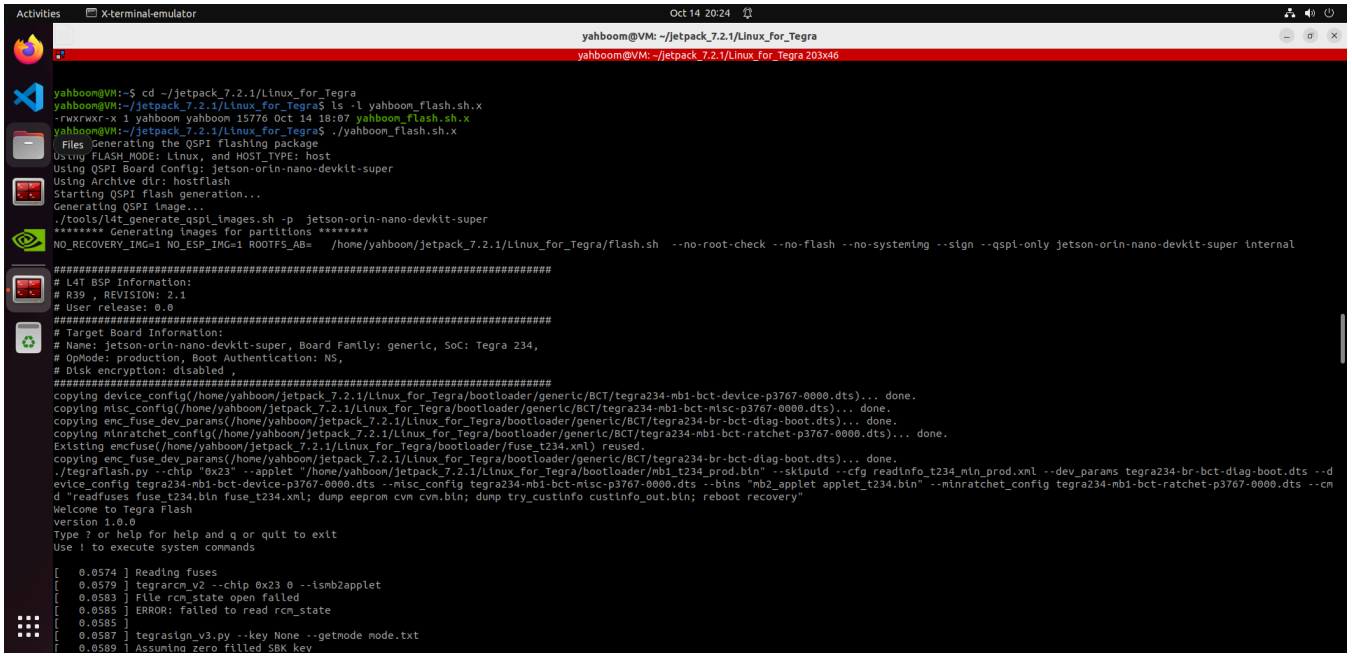
```
cd ~/Jetpack7.2.1/Linux_for_Tegra
ls -l yahboom_flash.sh.x
```

Then run:

```
./yahboom_flash.sh.x
```

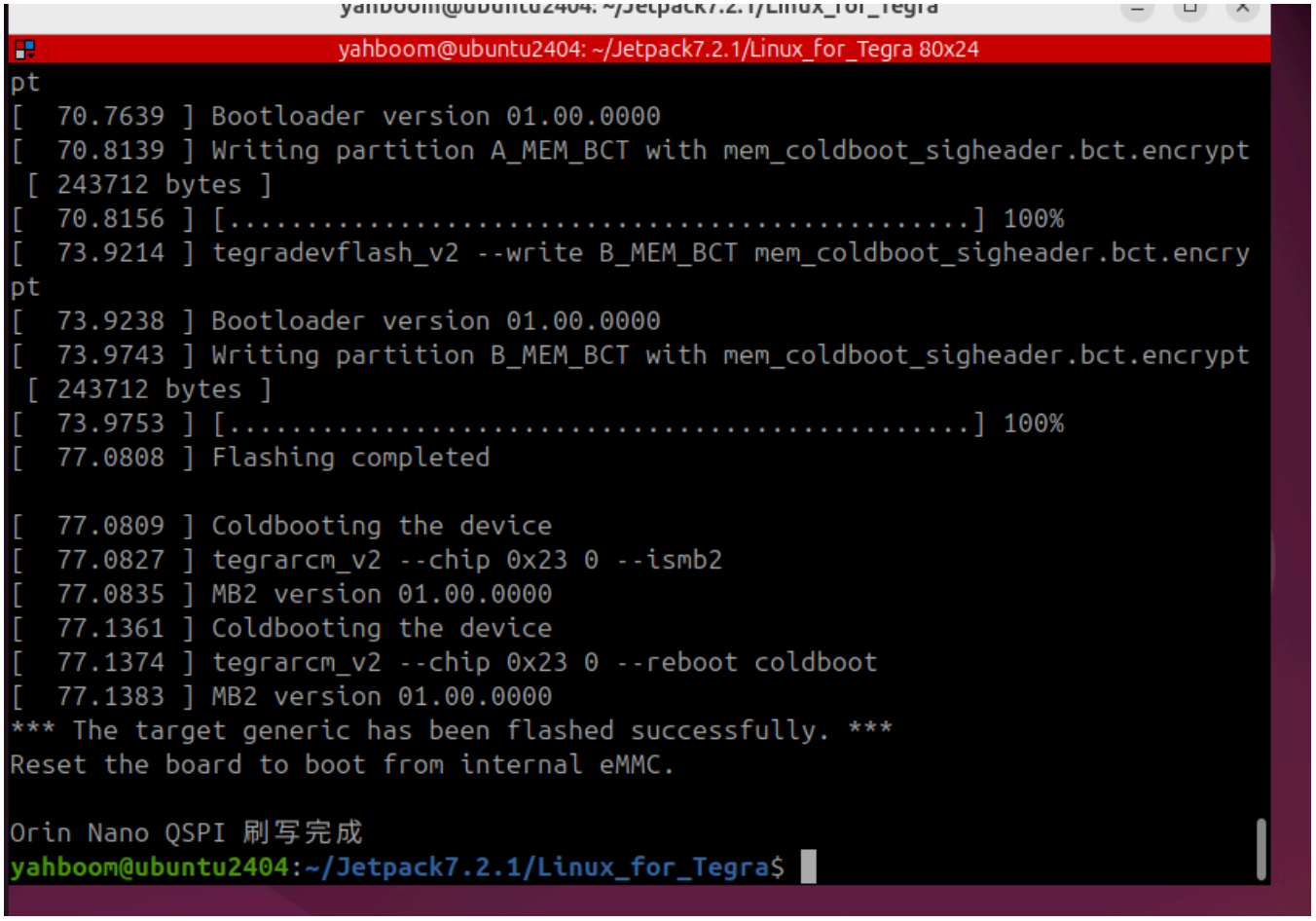
The script performs the following steps in order:

1. Generate the QSPI bootloader flashing package of R39.2.1.
2. Extract the generated flashing package.
3. Send the temporary flashing system to the mainboard, and the mainboard performs the bootloader write.



Once the following content appears, the flashing is complete

```
***The target generic has been flashed successfully***
Reset the board to boot from internal eMMC.
```



Note: At this point the NVIDIA APX device disconnects and the mainboard fan may start to spin. This is a mode switch that may occur when the temporary flashing system takes over; APX is not required to stay connected. Do not power off, force the mainboard back into Recovery mode or repeatedly reconnect APX because of it.

During the wait, no new log appears in the terminal, and the script used here does not reboot the mainboard by default either. After the wait is over and the script returns to the command prompt, continue with the operations in section 3.

If the script reports an error early on, first check the error message above it. After confirming that the process has stopped, power off, re-enter Recovery mode as described in section 1 and reconnect the virtual machine; do not power off while the board is still being written to.

3. Start the system

1. After the flashing script has finished, disconnect the Jetson power and the Type-C data cable. If a debug serial port is connected, first confirm that the write on the board has completed.
2. Remove the Recovery jumper cap that shorts `FC REC` and `GND`.
3. Install the bundled JetPack 7.2.1 SSD and connect a display.
4. Power the mainboard up again and watch the UEFI firmware version on the boot screen; it should show `39.2.1`.

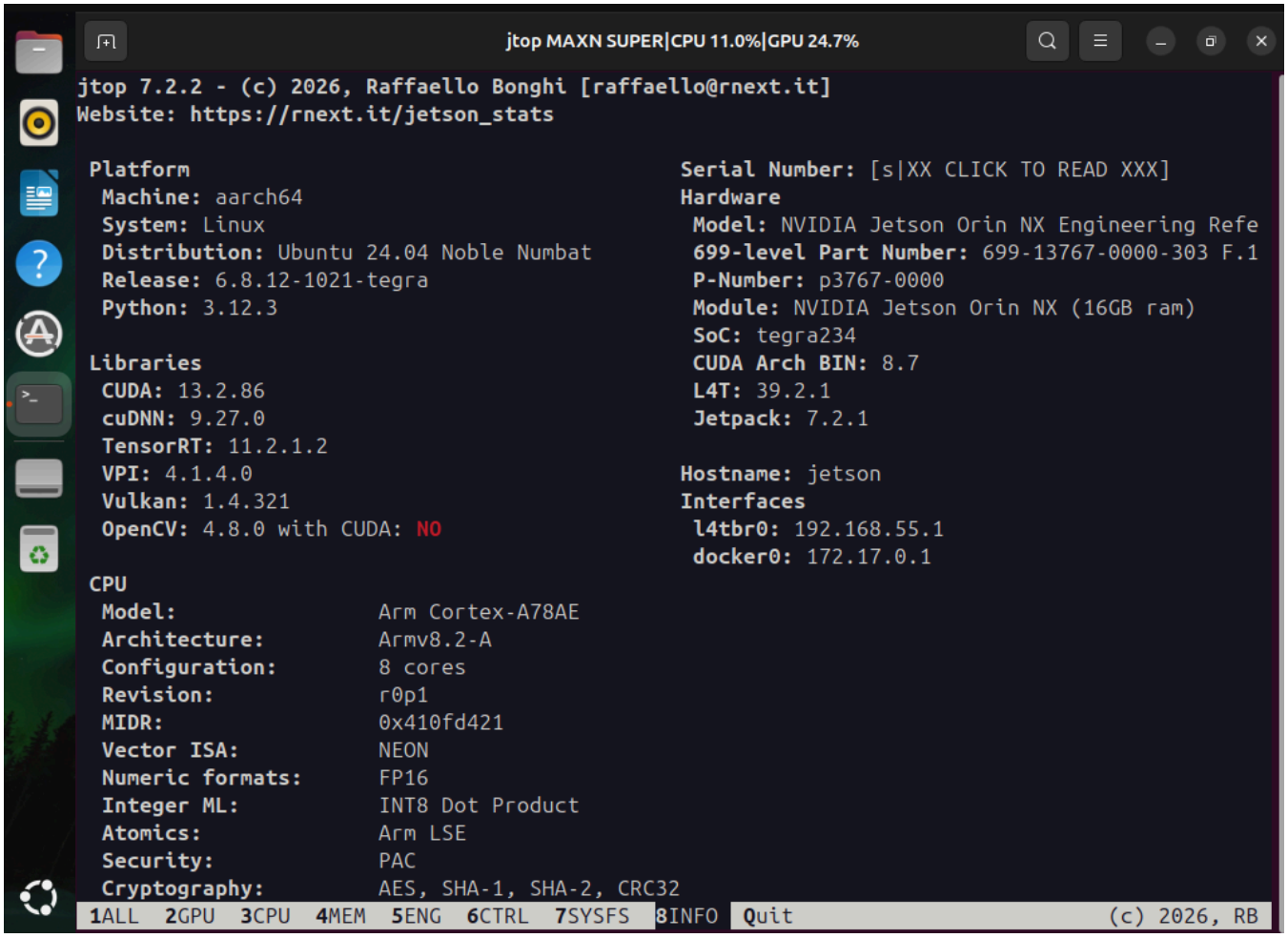


If it still shows `36.x`, the bootloader version acceptance of this run has not been completed; check whether the correct bundled files and the newly generated flashing package were used, and investigate further with the board-side log.

After entering the system, open a terminal and check further:

```
sudo nvbootctrl dump-slots-info
cat /etc/nv_tegra_release
```

The first command shows the bootloader version and the slot information of the mainboard; the second one shows the system version on the SSD, and it should show `R39`, `REVISION: 2.1`. **The SSD system version cannot replace the bootloader version check.**



When it is done, reboot normally once and confirm that the mainboard can boot from the bundled SSD; you can then continue with the following tutorials.